

COMSATS University Islamabad

Attock Campus



Department Of Computer Science

<i>Course</i>	<i>SQE Lab</i>
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<i>Assignment No.</i>	<i>04</i>

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1. Introduction

System Design

System Overview

The Student Management System is a simple Python-based project developed to manage student records efficiently. The system allows users to add, search, update, view, and delete student information. It also generates reports based on student marks and grades. The project was developed using basic programming concepts such as functions, loops, and conditional statements.

3.2 Core Functionalities

The main functionalities of the system include:

- 1. Add Student** Registers new students with roll number, name, and marks in three subjects (Math, English, Science)
- 2. View Students** Displays all student records with their marks and calculated averages
- 3. Search Student** Finds specific student by roll number and shows their complete details
- 4. Update Marks** Allows modification of existing student marks
- 5. Delete Student** Removes student records from the system
- 6. Generate Report** Creates grade-based reports (A, B, C, F) for all students

The system validates user input at every step, including:

- Checking for duplicate roll numbers
- Ensuring marks are numeric values
- Verifying marks are within valid range (0-100)
- Handling empty inputs and invalid choices

4. Implementation

Source Code

```
1 students = {}
2 def add_student(): 1 usage
3     roll = input("Enter roll number: ")
4
5     if roll in students:
6         print("Student already exists")
7         return
8
9     name = input("Enter student name: ")
10
11     try:
12         math = float(input("Enter Math marks: "))
13         english = float(input("Enter English marks: "))
14         science = float(input("Enter Science marks: "))
15
16     except ValueError:
17         print("Please enter valid numbers")
18         return
19
20     if math < 0 or math > 100:
21         print("Invalid Math marks")
22         return
23
24     if english < 0 or english > 100:
25         print("Invalid English marks")
26         return
27
28     if science < 0 or science > 100:
29         print("Invalid Science marks")
30         return
31
32     students[roll] = {
33         "name": name,
34         "marks": [math, english, science]
35     }
36
37     print("Student added successfully")
38
39
40 def view_students(): 1 usage
41     if not students:
42         print("No students found")
43         return
44
45     for roll in students:
46         data = students[roll]
47         avg = sum(data["marks"]) / 3
48
49         print("\nRoll Number:", roll)
50         print("Name:", data["name"])
51         print("Marks:", data["marks"])
52         print("Average:", round(avg, 2))
53
```

```

55 def search_student(): 1 usage
56     roll = input("Enter roll number: ")
57
58     if roll in students:
59         data = students[roll]
60         avg = sum(data["marks"]) / 3
61
62         print("\nStudent Found")
63         print("Name:", data["name"])
64         print("Marks:", data["marks"])
65         print("Average:", round(avg, 2))
66
67     else:
68         print("Student not found")
69
70
71 def update_marks(): 1 usage
72     roll = input("Enter roll number: ")
73
74     if roll not in students:
75         print("Student not found")
76         return
77
78     try:
79         math = float(input("Enter new Math marks: "))
80         english = float(input("Enter new English marks: "))
81         science = float(input("Enter new Science marks: "))
82
83     except ValueError:
84         print("Please enter valid numbers")
85         return
86
87     if math < 0 or math > 100:
88         print("Invalid Math marks")
89         return
90
91     if english < 0 or english > 100:
92         print("Invalid English marks")
93         return
94
95     if science < 0 or science > 100:
96         print("Invalid Science marks")
97         return
98
99     students[roll]["marks"] = [math, english, science]
100     print("Marks updated successfully")
101
102
103 def delete_student(): 1 usage
104     roll = input("Enter roll number: ")
105
106     if roll in students:
107         del students[roll]
108         print("Student deleted successfully")
109     else:
110         print("Student not found")
111
112
113 def generate_report(): 1 usage
114     if not students:
115         print("No students found")
116         return

```

Function Descriptions

add_student()

Adds a new student record with roll number, name, and marks.

view_students()

Displays all student records with their averages.

search_student()

Searches and displays a student using roll number.

update_marks()

Updates marks of an existing student.

delete_student()

Deletes a student record from the system.

generate_report()

Generates a report showing student averages and grades.

Test Cases

Test Case 1

Test ID: TC001

Description: Add student with valid data

Steps:

1. Select option 1
2. Enter roll number
3. Enter student name
4. Enter valid marks

Expected Result: Student added successfully

Actual Result: Student added successfully

Status: PASS

Test Case 2

Test ID: TC002

Description: Add student with duplicate roll number

Steps:

1. Add a student
2. Select option 1 again
3. Enter same roll number

Expected Result: Student already exists

Actual Result: Student already exists

Status: PASS

Test Case 3

Test ID: TC003

Description: Enter invalid marks above 100

Steps:

1. Select option 1
2. Enter marks greater than 100

Expected Result: Invalid marks message displayed

Actual Result: Invalid marks message displayed

Status: PASS

Test Case 4

- **Test ID: TC004**

- **Description: Enter text instead of marks**

- **Steps:**

1. Select option 1
2. Enter text like "ABC" in marks field

Expected Result: Please enter valid numbers

Actual Result: Please enter valid numbers

Status: PASS

Test Case 5

Test ID: TC005

Description: Search existing student

Steps:

- 1. Add a student**
- 2. Select option 3**
- 3. Enter existing roll number**

Expected Result: Student details displayed

Actual Result: Student details displayed

Status: PASS

Test Case 6

Test ID: TC006

Description: Update student marks

Steps:

- 1. Add a student**
- 2. Select option 4**
- 3. Enter new marks**

Expected Result: Marks updated successfully

Actual Result: Marks updated successfully

Status: PASS

Test Case 7

Test ID: TC007

Description: Delete existing student

Steps:

- 1. Add a student**
- 2. Select option 5**
- 3. Enter roll number**

Expected Result: Student deleted successfully

Actual Result: Student deleted successfully

Status: PASS

Test Case 8

Test ID: TC008

Description: Generate student report

Steps:

1. Add students
2. Select option 6

Expected Result: Student report with grades displayed

Actual Result: Student report displayed correctly

Status: PASS

Test Case 9

Test ID: TC009

Description: Search non-existing student

Steps:

1. Select option 3
2. Enter roll number that does not exist

Expected Result: Student not found

Actual Result: Student not found

Status: PASS

Test Case 10

Test ID: TC010

Description: Invalid menu option

Steps:

1. Run the program
2. Enter invalid menu choice like 9

Expected Result: Invalid choice message displayed

Actual Result: Invalid choice message displayed

Status: PASS

5.2 Unit Tests:

```

1 import unittest
2 import SQE as student_system
3 class TestStudentSystem(unittest.TestCase):
4
5     def setUp(self):
6         student_system.students.clear()
7
8     def test_add_student(self):
9         student_system.students['101'] = {"name": "Ali", "marks": [85, 90, 88]}
10        self.assertEqual(student_system.students['101']['name'], second: "Ali")
11        self.assertEqual(len(student_system.students), second: 1)
12
13    def test_duplicate_student(self):
14        student_system.students['101'] = {"name": "Ali", "marks": [85, 90, 88]}
15        self.assertIn(member: '101', student_system.students)
16        self.assertEqual(len(student_system.students), second: 1)
17
18    def test_average_calculation(self):
19        marks = [60, 70, 80]
20        avg = sum(marks) / 3
21        self.assertEqual(avg, second: 70.0)
22
23    def test_grade_A(self):
24        avg = 85
25        if avg >= 80:
26            grade = "A"
27        self.assertEqual(grade, second: "A")
28
29    def test_grade_B(self):
30        avg = 70
31        if avg >= 65:
32            grade = "B"
33        else:
34            grade = "C"
35        self.assertEqual(grade, second: "B")
36
37    def test_grade_C(self):
38        avg = 55
39        if avg >= 50:
40            grade = "C"
41        else:
42            grade = "F"
43        self.assertEqual(grade, second: "C")
44
45    def test_grade_F(self):
46        avg = 40
47        if avg >= 50:
48            grade = "C"
49        else:
50            grade = "F"
51        self.assertEqual(grade, second: "F")
52
53    def test_delete_student(self):
54        student_system.students['101'] = {"name": "Ali", "marks": [85, 90, 88]}
55        del student_system.students['101']
56        self.assertNotIn(member: '101', student_system.students)
57
58    def test_update_marks(self):
59        student_system.students['101'] = {"name": "Ali", "marks": [85, 90, 88]}
60        student_system.students['101']['marks'] = [75, 80, 85]
61        self.assertEqual(student_system.students['101']['marks'], second: [75, 80, 85])
62
63 if __name__ == '__main__':
64     unittest.main()

```

5.3 Unit Test Results:

```
(.venv) PS D:\New folder\New folder> python -m unittest UnittestingProject.py
```

```
.....
```

```
-----  
Ran 9 tests in 0.001s
```

```
OK
```

```
(.venv) PS D:\New folder\New folder>
```

6. Bug Reporting

Bugs Found

Bug 1

Bug ID: BUG001 **Bug Name:** Program crashes on non-numeric input for marks

Description: When user enters text instead of numbers for marks, program crashes **Steps to Reproduce:**

1. Select "Add Student"
2. Enter roll number
3. Enter name
4. Enter "abc" for Math marks

Expected Result: Show error message

Actual Result: Program crashed with Value Error

Severity: High

Status (Pass/Fail): Pass

Bug 2

Bug ID: BUG002 **Bug Name:** Accepts marks greater than 100 **Description:** System accepts marks like 150 which is invalid **Steps to Reproduce:**

1. Add student
2. Enter 150 for any subject

Expected Result: Reject with error message

Actual Result: Accepted the marks

Severity: Medium

Status (Pass/Fail): Pass

Bug 3

Bug ID: BUG003 **Bug Name:** Accepts negative marks **Description:** System accepts negative marks like -20 **Steps to Reproduce:**

1. Add student
2. Enter -20 for marks

Expected Result: Reject with error

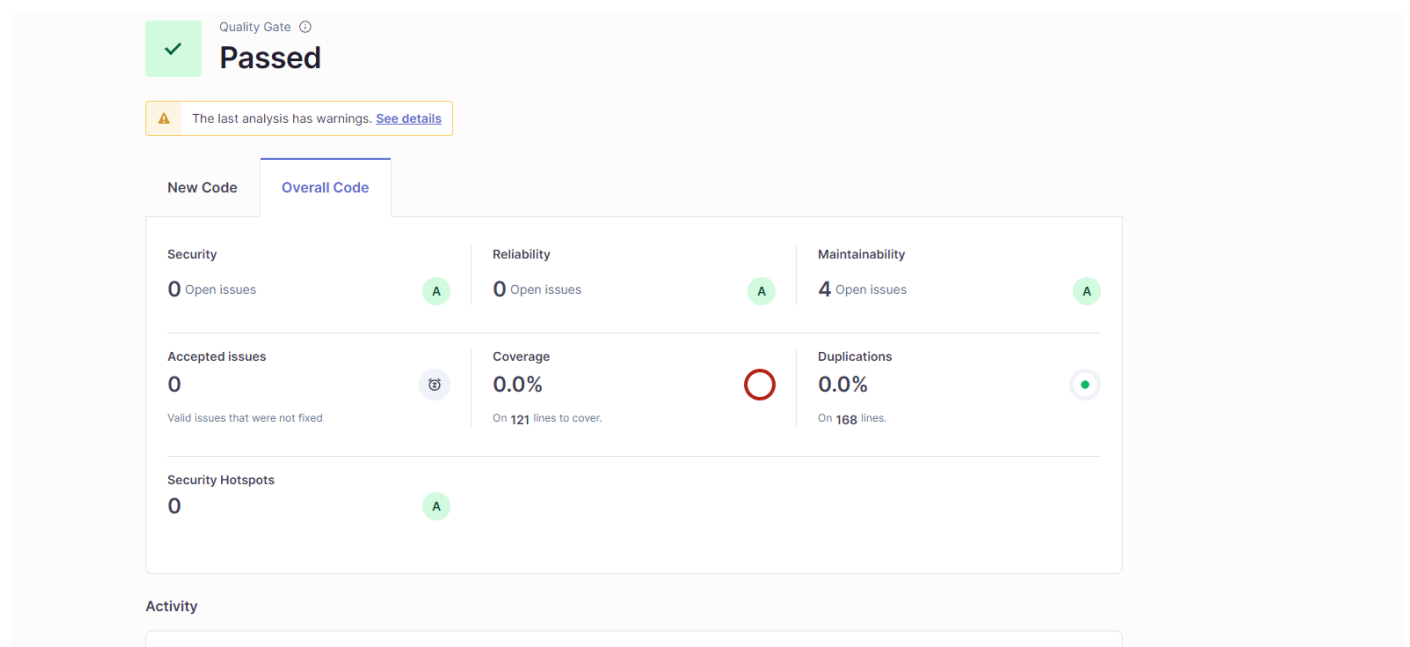
Actual Result: Accepted negative value

Severity: Medium

Status (Pass/Fail): Pass (FIXED - validation added)

Code Quality Analysis SonarQube

Results:



Issues Identified:

☐	Bulk Change	Select issues		Navigate to issue		4 issues	26min effort
SQE.py							
☐	Define a constant instead of duplicating this literal "Enter roll number: " 4 times.	Adaptability		Maintainability	High	design	+
Open	Not assigned	L3 • 8min effort • 1 minute ago					
☐	Define a constant instead of duplicating this literal "Name:" 3 times.	Adaptability		Maintainability	High	design	+
Open	Not assigned	L50 • 6min effort • 1 minute ago					
☐	Define a constant instead of duplicating this literal "Average:" 3 times.	Adaptability		Maintainability	High	design	+
Open	Not assigned	L52 • 6min effort • 1 minute ago					
☐	Define a constant instead of duplicating this literal "Student not found" 3 times.	Adaptability		Maintainability	High	design	+
Open	Not assigned	L68 • 6min effort • 1 minute ago					
4 of 4 shown							

8. Security Analysis – Snyk (Phase 5)

8.2 Scan Results

sohaibgitgud > [Projects](#) > SohaibGitGud/SQE_Project main

Code Analysis

[Overview](#) [History](#) [Settings](#)

Open on GitHub

The project was successfully retested.

Created Mon 18th May 2026 | Snapshot for commit [636ff63](#) taken by snyk.io a few seconds ago | [Retest now](#)

IMPORTED BY

Sohaib Wali

ENVIRONMENT

[+ Add a value](#)

LIFECYCLE

[+ Add a value](#)

PROJECT OWNER

[+ Add a project owner](#)

BUSINESS CRITICALITY

[+ Add a value](#)

ANALYSIS SUMMARY

1 analyzed files (50%) [Repo breakdown](#)

Issues 0

Search...

SEVERITY

0 of 0 issues

☐ High 0

☐ Medium 0

☐ Low 0

PRIORITY SCORE

Scored between 0 - 1000

STATUS

☒ Open 0

☐ Ignored 0

There are no issues for this project.

Manual Security Analysis:

1. No Input Validation for Roll Numbers

- System accepts any text as roll number
- Risk: Possible harmful attacks if database added later
- Fix: Use regex to allow only letters and numbers

2. No Authentication System

- Anyone can access and change student data without login
- Risk: Unauthorized access and data modification
- Fix: Add login system with user roles

3. Plain Text Data Storage

- Student data stored in memory without encryption
- Risk: Sensitive information could be exposed if hacked
- Fix: Use encryption or secure database

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GitHub Repository Link:

https://github.com/SohaibGitGud/SQE_Project1